ONTI · ONTOLOGY ANALYSIS REPORT

VNB Ontology Analytics — Test Report

Generated: 2026-10-27T08:24:57+00:00
Ontology pack: VNB / APE Insurance Sample Pack

2,025 Business Year	**$7.42M** Stated Total APE	**$1.45M** Stated Total VNB
19.5% Stated VNB Margin	**$7.42M** Derived Total APE	**$1.45M** Derived Total VNB

Date: 2026-10-27 | **Ontology pack:** VNB / APE Insurance Sample Pack | **Scope:** Session | **Questions analysed:** 3

1. Executive Summary

- Ontology context — How the ontology explains the business The knowledge graph connects a top-level financial KPI (**Total VNB**) to operational drivers (agents, productivity, churn) and product economics (mix weights, product margins) through **31 directed edges of 4 types**.
- Analysed **3 question(s)** (3 rule-engine, 0 AI-assisted).
- Key figures: **Business Year 2,025**, **Stated Total APE $7.42M**, **Stated Total VNB $1.45M**, **Stated VNB Margin 19.5%**.
- Highest Unit-Linked at 25.0%; lowest Term Life at 10.0%. Grouped bars compare VNB Margin, Commission Rate, Expense Ratio.
- Highest item 2 at 66.0%; lowest item 4 at 5.0%. Grouped bars compare Mix Weight, Year.

2. Q&A Record

1. What is the total VNB?

*Mode: **Rules Engine** · 12.4 ms*

The stated **Total VNB is $11.08M** on APE of **$68.00M**.

Metric	Value
APE	$68.00M
VNB	$11.08M

Ontology evidence: TotalVNB, TotalAPE, BlendedVNBMargin

2. How many channels are there and what do they contribute?

*Mode: **Rules Engine** · 18.9 ms*

Three distribution channels:

- **Agency** — APE $20.00M, VNB $3.62M
- **Bancassurance** — APE $30.00M, VNB $6.55M
- **Brokerage** — APE $18.00M, VNB $3.10M

Ontology evidence: ChannelAPE, TotalAPE, AgentCount, AgentProductivity

3. Why does the blended margin differ across channels?

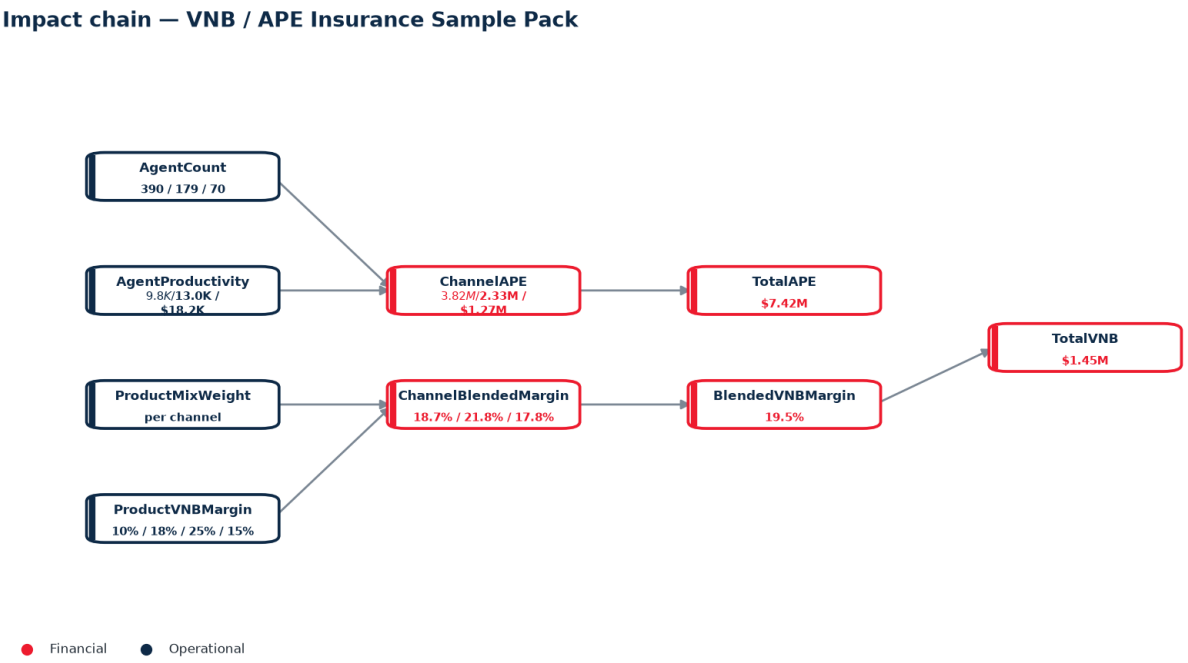
*Mode: **Rules Engine** · 9.1 ms*

Blended margin is driven by **product mix weights** and per-product VNB margins; Banca leans on Participating products (21.85%).

Ontology evidence: ProductMixWeight, ProductVNBMargin, ChannelBlendedMargin, BlendedVNBMargin

3. Ontology Evidence — Impact Chain

The diagram below shows the directed reasoning chain linking operational drivers to KPI outcomes captured by the analysed questions.



Impact chain — VNB / APE Insurance Sample Pack

Entity & attribute definitions

Entity	Layer	Attributes
VNB Monthly Results	1	Month, Total APE, Total VNB, VNB Margin, Agency APE, Agency VNB, Agency share of VNB, MoM VNB change
Channel Monthly KPIs	2	Month, Channel ID, Active agents, Avg productivity, APE, Weighted VNB margin, VNB
Agent Monthly Performance	3	Month, Agent ID, Channel ID, Status, Productivity, APE, Churn risk
Product Mix Monthly	3	Month, Channel ID, Product ID, Product mix, VNB margin, Weighted margin component
Scenario Analysis	4	Scenario / input, Value, Unit / interpretation

Metric formulas (calculation basis)

Metric	Formula
Business Year	`cell('kpi', 'year')`
Stated Total APE	`cell('kpi', 'ape')`
Stated Total VNB	`cell('kpi', 'vnb')`
Stated VNB Margin	`cell('kpi', 'margin')`
Derived Total APE	`round(sum(rows('channels'), 'ape'), 2)`

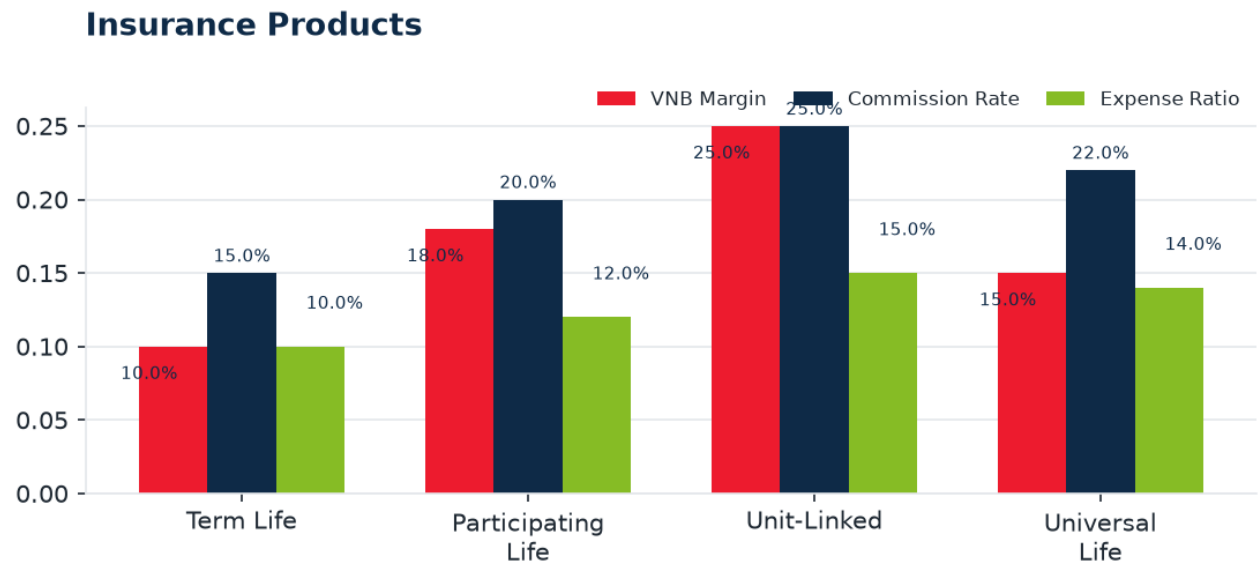
Metric	Formula
Derived Total VNB	`round(sum(rows('channels'), 'vnb'), 3)`
Derived VNB Margin	`round(derived_vnb / derived_ape, 4)`
VNB Gap	`round(derived_vnb - stated_vnb, 2)`
Margin Gap	`round(derived_margin - stated_margin, 4)`

4. KPI and Numbers

Metric	Value	Unit
Business Year	**2,025**	
Stated Total APE	**$7.42M**	$M
Stated Total VNB	**$1.45M**	$M
Stated VNB Margin	**19.5%**	%
Derived Total APE	**$7.42M**	$M
Derived Total VNB	**$1.45M**	$M
Derived VNB Margin	**19.5%**	%
VNB Gap	**$-0.00M**	$M
Margin Gap	**0.0%**	%

5. Chart Analysis

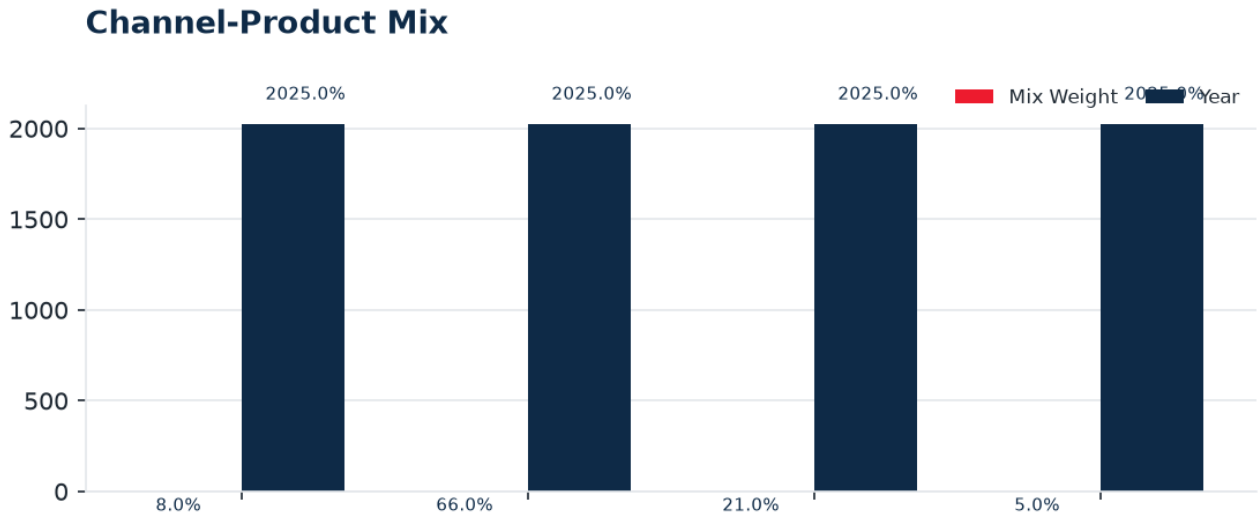
1. Insurance Products



Insurance Products

Highest **Unit-Linked** at 25.0%; lowest **Term Life** at 10.0%. Grouped bars compare VNB Margin, Commission Rate, Expense Ratio.

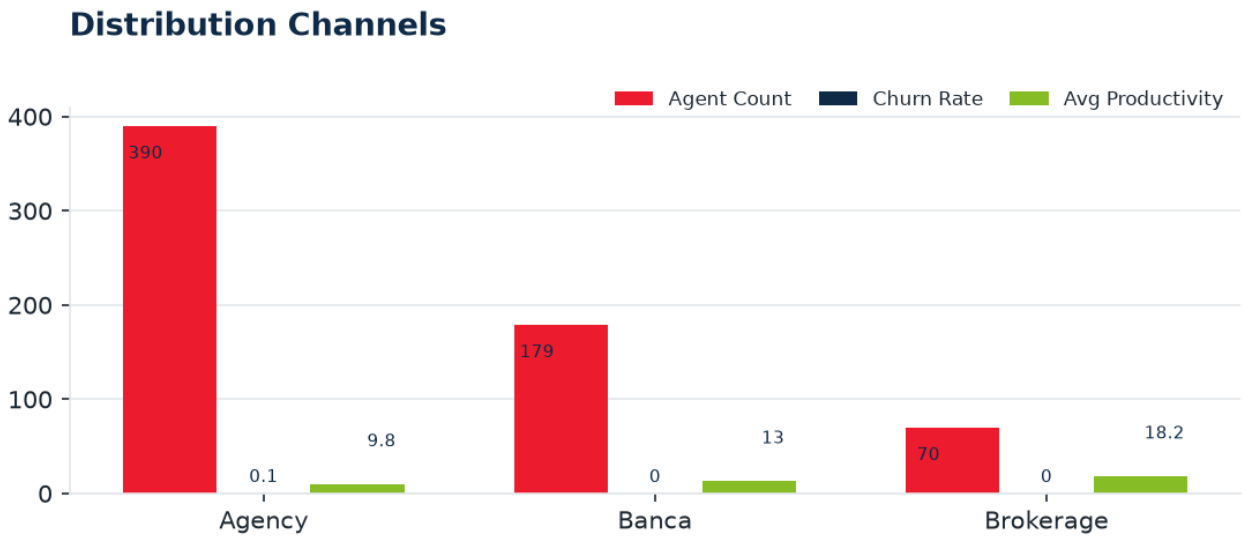
2. Channel-Product Mix



Channel-Product Mix

Highest **item 2** at 66.0%; lowest **item 4** at 5.0%. Grouped bars compare Mix Weight, Year.

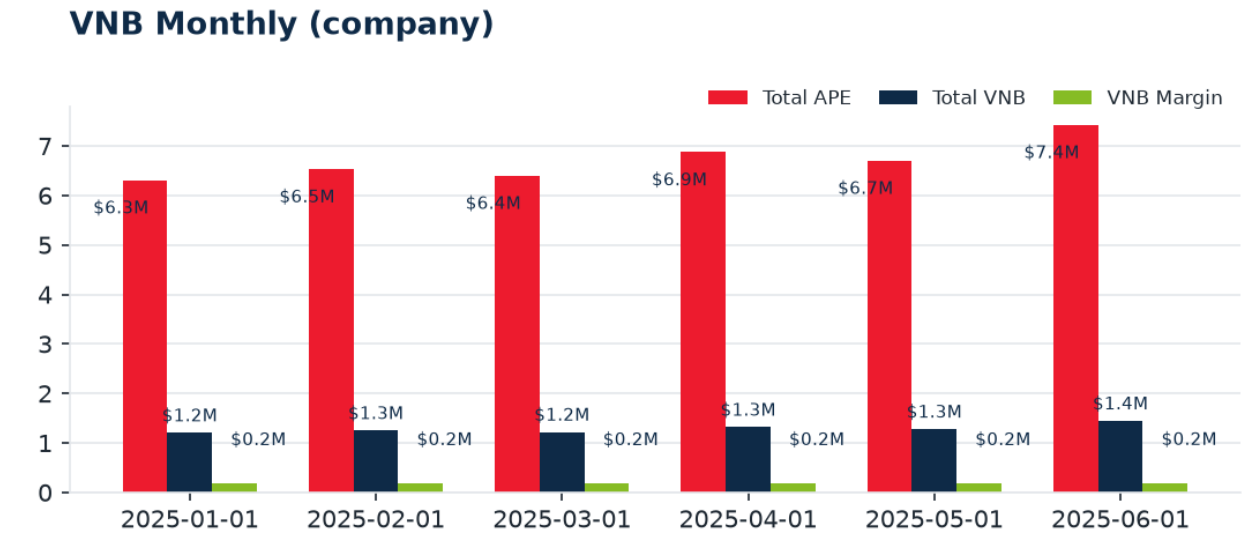
3. Distribution Channels



Distribution Channels

Highest **Agency** at 390; lowest **Brokerage** at 70. Grouped bars compare Agent Count, Churn Rate, Avg Productivity.

4. VNB Monthly (company)



VNB Monthly (company)

Highest **2025-06-01** at $7.4M; lowest **2025-01-01** at $6.3M. Grouped bars compare Total APE, Total VNB, VNB Margin.

6. Appendix

Methodology & assumptions

- Answers are produced by Onti's rule engine against the active ontology pack; AI answers are grounded with the same knowledge base and must not introduce external facts.
- Derived columns and metric formulas are evaluated in declaration order from the active dataset; the calculation basis is listed in the Ontology Evidence section.
- Charts render only tables with 2-15 rows in their active dataset; sections are omitted when their source data is absent.

Dataset inventory

Table	Entity	Active dataset	Rows
Company KPI (year)	vnb_summary	Seed: Company KPI (year)	1
Insurance Products	insurance_product	Seed: Insurance Products	4
Channel-Product Mix	channel_product_mix	Seed: Channel-Product Mix	4
Distribution Channels	distribution_channel	Seed: Distribution Channels	3
Agents	agent	Seed: Agents	500
VNB Monthly (company)	vnb_monthly	Seed: VNB Monthly (company)	6
Channel Monthly (KPIs)	channel_monthly	Seed: Channel Monthly (KPIs)	18
Product Mix Monthly	product_mix_monthly	Seed: Product Mix Monthly	24
Agent Master	agent	Seed: Agent Master	500
Agent Monthly Performance	agent_monthly	Seed: Agent Monthly Performance	3000
KPI Linkage (ontology)		Seed: KPI Linkage (ontology)	7
Scenario Analysis (seed)	scenario_analysis	Seed: Scenario Analysis (seed)	15

Generation metadata

Item	Value
Report ID	`36139f9a62e5`
Generated at (UTC)	2026-10-27T08:24:57+00:00
Sections included	summary, qa, evidence, kpi, charts, appendix
AI interpretation	off